

ORIGINAL ARTICLE

Large language models for efficient topic modeling

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Abstract

The utilization of large language models (LLMs) in research is becoming increasingly prevalent, as they offer advanced capabilities in processing and generating human-like text. However, this advancement comes with a significant trade-off in terms of time and computational costs. In this paper, we demonstrate that analyzing large text datasets with the use of LLMs can be performed efficiently in terms of both time and energy. For this purpose, we utilize the Llama pre-trained model. In more detail, we study the topic modeling task where the goal is to discover and identify topics in large text corpora. The basis of our approach is a hierarchical divisive clustering technique that clusters the data based on their semantic similarity, after employing a Sentence-BERT encoder, pre-trained on a variety of data across different tasks. Then, using an LLM, we identify topics for representative samples from each cluster. Additionally, we introduce a new evaluation method that leverages the capabilities of LLMs to assess the alignment between discovered topics and ground truth labels, providing a robust validation metric. Our findings indicate that it is possible to effectively reduce the computational cost of the topic modeling process compared to the direct application of LLMs and BERTopic, while simultaneously enhancing inference time and overall efficiency, thereby surpassing the current state-of-the-art capabilities of BERTopic.

Keywords Large language models · Hierarchical divisive clustering · Llama 3.1 · BERTopic

1 Introduction

In today's era of information abundance, the ability to extract meaningful insights from vast text collections has become crucial. However, raw text data in its unstructured form are not inherently useful, necessitating sophisticated tools and techniques to distill relevant semantic information. Topic modeling, which integrates machine learning (ML) and natural language processing (NLP), plays a pivotal role in uncovering latent topics and patterns within unorganized text. By organizing and summarizing large corpora, topic modeling enables efficient exploration of data and the extraction of valuable insights [1].

At its core, topic modeling is an unsupervised ML technique that identifies underlying thematic structures within a large corpus by analyzing word co-occurrences and frequencies. Early techniques in this field relied on statistical methods to detect groups of words that best summarized key information in a document collection [2]. These approaches aimed to represent documents as mixtures of topics, where each topic is characterized by a distribution of words. This allows for the automatic inference of topics based on the patterns of word usage across documents, revealing hidden semantic themes that would otherwise remain obscured [3].

A key breakthrough in topic modeling came with the introduction of latent Dirichlet allocation (LDA), one of the most influential models in this domain. LDA, proposed by Blei et al. in 2003, is a generative probabilistic



model that represents each document as a mixture of topics and each topic as a distribution of words. This unsupervised method has been widely adopted for its ability to automatically discover semantically coherent topics from large corpora, without requiring prior knowledge of topics or labels [2]. As a result, LDA has become a foundational technique for tasks such as document classification, search and recommendation systems.

Over time, topic modeling has evolved with the advancement of deep learning. Neural topic modeling leverages Deep Learning techniques, allowing models to learn relationships between topics and words through back-propagation. This approach enhances traditional topic modeling by capturing more complex patterns and relationships, without the need for human-annotated training data [4]. In parallel, the development of the transformer architecture, powered by the attention mechanism, has revolutionized NLP by enabling models to process entire contexts simultaneously [5]. This has given rise to Large Language Models (LLMs), characterized by their vast number of parameters and extensive training on large datasets. LLMs demonstrate remarkable language understanding and have transformed core NLP tasks, such as sentiment analysis, entity recognition, and topic modeling [6].

If we now turn our attention to the environmental impacts of modern Artificial Intelligence (AI) systems, we notice that they are often described as high energy-consuming systems [7]. Actions have been taken by many organizations across the world to address this issue. Most recently, the 28th United Nations Climate Change Conference (COP28) acknowledged the imperative for deep and rapid reductions in greenhouse gas emissions [8]. Additionally, the European Union has committed to reducing energy consumption by 11.7% by 2030 [9]. It is widely understood that the continued evolution of AI systems must comply with and strive for the lowest possible environmental impact.

Despite the exceptional ability of LLMs to mimic complex language use, their implementation poses notable environmental challenges. The massive scale and intricate architectures of LLMs demand substantial computational resources, often resulting in high energy consumption for model training and inference. Recent estimates suggest that training an LLM like GPT-3 can generate carbon emissions equivalent to the lifetime footprint of nearly five passenger vehicles [10]. The intensive processing power required by LLMs has raised concerns about associated greenhouse gas emissions and overall energy demands, particularly if the use of these models scales up dramatically [11].

In this work, we introduce an energy and time-efficient approach for unsupervised topic modeling of large-scale text corpora. Our approach employs a computationally efficient divisive hierarchical clustering method to first segment documents based on semantic similarity. We sample a fraction of exemplars from each distinct cluster and utilize pre-trained LLMs' ability to understand and discover topics at the cluster level. By transferring these topics to all documents in each cluster, we can efficiently annotate massive datasets without needing to pass each document individually through the LLM. Through targeted sampling and transfer of cluster-level topics, we avoid costly runtime and resources by inducing topics from stratified exemplars rather than annotating the full corpora with LLMs. We quantitatively demonstrate improvements in time and energy efficiency over baseline methods, while maintaining competitive topic modeling quality on established benchmarks. Additionally, we introduce an LLM-based evaluation method to assess the alignment between discovered topics and ground truth labels, providing thorough validation of our method's effectiveness. Notice that we aim to identify latent thematic structures in unannotated textual data through unsupervised topic modeling, without any predefined topic categories. The main contributions of this research are outlined as follows:

- We propose an energy and time-efficient approach for unsupervised topic modeling of large text corpora that has improved efficiency over traditional baseline methods while maintaining competitive quality in topic modeling.
- We show that we can reduce the need for expensive LLM inference on each document, thereby saving time and computational resources by transferring cluster-level topics to all documents within each cluster.
- We introduce a new LLM-based evaluation method that quantitatively assesses the alignment between discovered topics and ground truth labels, providing a validation approach for unsupervised topic modeling.

To show the applicability of the proposed approach, we report a comparison analysis of other state-of-the-art methodologies in terms of induced topic precision and energy efficiency. The rest of the manuscript is organized as follows, Sect. 2 provides the related literature. Section 3 outlines the proposed framework, including details on the used data, the divisive hierarchical clustering, and the LLM setup. Section 4 presents the experimental outputs of our framework on multiple datasets, followed by Sect. 5 that analyzes the results, specifically interpreting model behaviors, as well as the framework's advantages and limitations. We also discuss potential applications and directions for future work. Finally, Sect. 6 summarizes the key conclusions, highlighting the main contributions toward efficient and scalable topic modeling through our ground-breaking combination of semantic clustering and LLM capabilities.

2 Related work

Topic modeling is a fundamental task in NLP that aims to automatically discover abstract topics within a collection of documents. The primary goal is to identify coherent groups of semantically related words that represent underlying themes, enabling better document understanding, organization, and retrieval. The literature on topic modeling has evolved from statistical approaches to increasingly sophisticated neural and LLM-based methods. Traditional topic modeling approaches like latent Dirichlet allocation (LDA) and its variants have been fundamental in discovering latent topics in document collections. However, these methods often struggle with limited semantic understanding and coherence. A comparative analysis highlighted these limitations, demonstrating that even basic matrix decomposition techniques like Non-negative Matrix Factorization (NMF) could produce more coherent topics in certain domains [12]. Several specialized approaches have been developed to address the challenges of short text analysis. The Biterm Topic Model (BTM) [13] introduced a novel way to handle short texts by modeling word co-occurrence patterns at the corpus level rather than the document level. Building on this foundation, [14] proposed ENNETM, which combines Gaussian and Logistic coding networks with pre-trained word embeddings to improve topic extraction performance and interpretability in short texts. A significant advancement in modern topic modeling came with BERTopic [15], which leverages transformer-based embeddings and introduces a class-based variant of TF-IDF for creating topic representations. This approach demonstrates superior performance in both topic coherence and diversity across different datasets, establishing a new benchmark for topic modeling quality.

While traditional approaches like LDA provide interpretable topic-word distributions, they often fail to capture semantic relationships between words. Neural approaches address this limitation but may sacrifice interpretability. Recent LLM-based methods offer both semantic understanding and interpretability but require significant computational resources. Recent advances in neural topic modeling have led to more sophisticated approaches. The Spherical Embedded Topic (SET) model [16] addresses key limitations of existing neural topic models by incorporating word semantic knowledge through von Mises-Fisher distributions and using a Wasserstein Auto-Encoder to prevent posterior collapse. Similarly, Lei et al. [17] introduced NMFT-LTM to address semantic misalignment in lifelong topic modeling through non-negative matrix tri-factorization.

The emergence of LLMs has opened new possibilities in topic modeling. Recent work by [18] demonstrates that LLMs can serve as an alternative to traditional topic modeling approaches, offering improved semantic understanding and topic coherence. Pham et al. [19] introduced TopicGPT, a prompt-based framework that produces more interpretable topics with natural language labels. Building on these advances, Xu et al. [20] proposed DeTiME, combining encoder-decoder LLMs with diffusion models to generate both coherent topics and topic-based text.

For handling document networks and graph structures, Zhang et al. [21] developed a GNN-based neural topic model that maintains interpretability while leveraging network connectivity. This approach uses the Optimal Transport Barycenter theory to capture network structure and semantic relationships between documents. Another significant advancement comes from [22], who introduced GCTM, addressing sample bias through graph-based sampling strategies and document-word information propagation.

The challenge of hierarchical topic modeling has also received attention, with [23] introducing HDDM, a variational autoencoder framework that learns topic hierarchies without requiring predefined structures. This approach demonstrates the importance of capturing hierarchical relationships in topic modeling while maintaining computational efficiency. Recent work has also focused on ensemble approaches to improve topic model stability and quality. Belford and Greene [24] proposed an ensemble method incorporating stability information through term co-associations and semantic similarity from word embeddings, demonstrating improved topic coherence and interpretability.

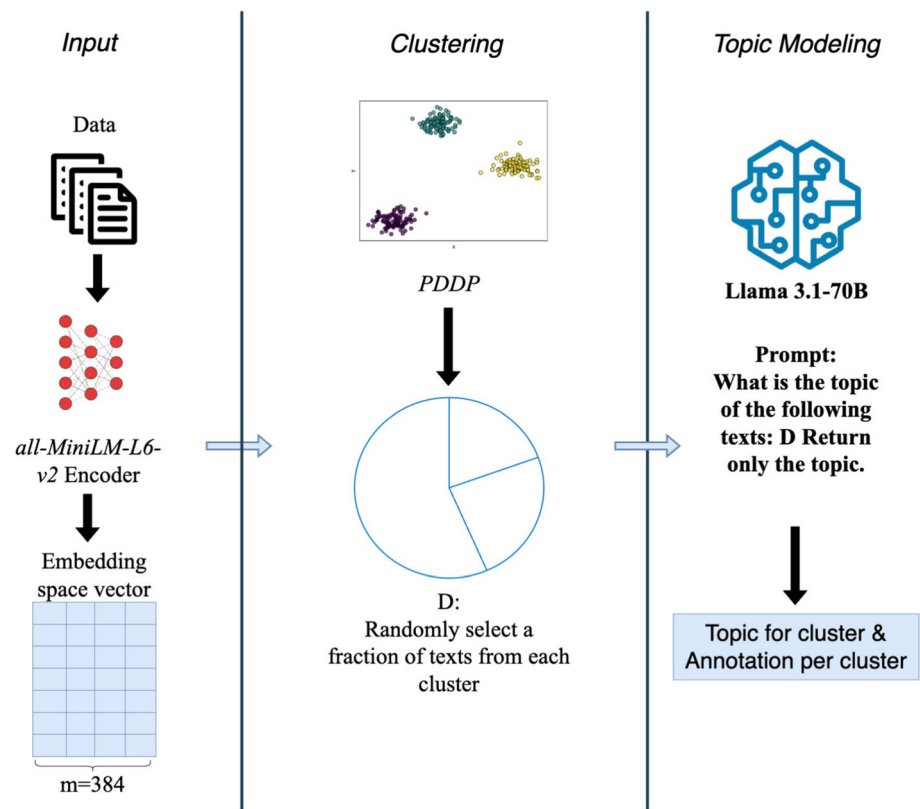
3 Research methodology

In this section, we provide the details of the proposed approach for efficient topic modeling of large text collections. The scope of this methodology encompasses the design decisions, algorithm details, modeling approaches, and evaluation procedures undertaken. The overall objective is to develop a topic modeling pipeline that can accurately annotate massive document collections with only minimal computational and energy requirements. To achieve this goal of efficiency without sacrificing modeling capabilities, we employ an algorithmic scheme that can be outlined as follows:

- The text data are transformed into an embedding space using the *all-MiniLM-L6-v2* Sentence-BERT encoder, pre-trained on a variety of data across different tasks. This step produces 384-dimensional dense vector representations for each text.
- A clustering algorithm, specifically Principal Direction Divisive Partitioning (PDDP), recursively partitions the data to form clusters based on semantic similarity. This hierarchical divisive approach allows for the efficient handling of large datasets. We utilized the PDDP algorithm implementation from the HiPart package, which provides an open-source native Python implementation of efficient and interpretable divisive hierarchical clustering algorithms [25].
- A sample of n texts is randomly selected from each cluster to serve as representative documents.
- The n representative texts from each cluster are processed by an LLM, specifically Llama 3.1 (70B parameter version Instruct), using a series of carefully engineered prompts to identify the topic related to each cluster.
- The knowledge gained from analyzing these representative samples is then efficiently transferred to all documents within the respective clusters, allowing for comprehensive topic annotation without the need to process every document individually through the LLM.

The architecture of our overall framework, including both the topic modeling pipeline and the evaluation method, is presented in Fig. 1, while the pseudocode is presented in Algorithm 1.

Fig. 1 Schematic overview of the proposed framework. This diagram illustrates the proposed framework, which consists of the following stages: (1) transformation of text corpora into an embedded space, (2) clustering using the PDDP algorithm, followed by selection of a representative fraction of documents from each cluster; and (3) application of a back-to-back prompting approach to Llama 3.1, enabling the discovery of topics within each cluster



Algorithm 1 Proposed Framework for Scalable Topic Discovery with Clustering and LLM Prompting

- 1: **Input:** Large collection of text documents
- 2: Transform documents into embedding space using *all-MiniLM-L6-v2* encoder
- 3: Select the number of clusters to separate the documents and ensuring their over-clustering
- 4: Cluster the embeddings using HiPart's PDDP algorithm
- 5: **for** each cluster **do**
- 6: Randomly select n representative texts
- 7: Prompt Llama LLM to identify the topic for the selected texts
- 8: Transfer identified topic to all documents in the cluster
- 9: **end for**
- 10: **Output:** Topic annotations for all documents

3.1 Text representation

Text representation is a crucial step in NLP. The main goal is to transform unstructured text data into a structured, machine-readable format. The purpose of text representation is to enable NLP algorithms to perform various tasks, such as text classification, sentiment analysis, and machine translation. The processed data has been transformed into contextualized representations. Contextualized representations refer to the ability of language models to generate unique, dynamic representations of words and sentences based on the surrounding context [26]. These representations not only capture the inherent meanings of individual words but also the small differences and subtleties of their usage within a specific context. The use of contextualized representations enhances the accuracy and naturalness of language output. It is increasingly vital for NLP tasks, such as sentiment analysis and question answering. Contextualized representations are achieved through Deep Neural Network architectures, trained on large text datasets. These models usually employ recurrent or transformer-based

architectures that can capture long-range dependencies between words and phrases in text. In this study, we transformed the text data into a 384-dimensional dense vector using the *all-MiniLM-L6-v2* model. This model is a Sentence-BERT encoder, pre-trained on a variety of data across different tasks [27]. This model transforms our text data into 384-dimensional dense vectors, providing rich, context-aware embeddings that serve as the foundation for subsequent analysis.

3.2 Clustering

With our text data transformed into high-dimensional vector representations, the next step is to identify meaningful patterns and groupings within this structured data space. For this purpose, we employed the Principal Direction Divisive Partitioning (PDDP) algorithm as our core method to recursively divide the text embeddings into clusters containing semantically similar texts. In this study, the PDDP algorithm implementation found in the HiPart package was utilized [25].

PDDP is a top-down hierarchical clustering technique that takes an innovative approach to split criteria. Unlike traditional methods, which use distance matrices, PDDP leverages the Principal Component Analysis (PCA) to identify the internal structural axes along which the data exhibits maximum variance. The clusters are then divided along this principal component direction to optimize the homogeneity of the resulting subgroups. By recursively splitting clusters along these locally determined principal axes, PDDP can reveal meaningful latent structures that more efficiently capture the semantic and thematic patterns within the text. Additionally, this localized covariance-based approach allows PDDP to scale efficiently to large datasets.

The operational workflow of PDDP begins by identifying the principal direction through eigenvalue decomposition on the covariance matrix of the data. Subsequently, the dataset is partitioned along these principal directions, forming distinct clusters. This process is repeated until the termination criterion is met. Unlike other hierarchical techniques, PDDP is completely deterministic, optimizes the variance at each split, and scales well to large datasets by concentrating on local cluster covariances [28, 29].

The distinctive feature of the PDDP algorithm is its minimal need for parameterization. Due to the nature of the selection and splitting criteria, there are no parameters to tune in these aspects. The only parameterization required involves setting the termination criterion. Consequently, the tunable parameters are the maximum number of splits to perform on the data and the minimum number of samples required in each cluster. Section 4 provides an analysis of these parameters.

3.3 Large language models and prompt engineering

The clusters identified by the PDDP algorithm form the foundation of our time and energy-efficient framework for large-scale text analysis. Instead of processing every document individually with an LLM, which would be computationally expensive, we leverage the semantic structure revealed by our clustering to strategically sample representative documents from each cluster. These carefully selected samples serve as proxies for their respective clusters, allowing us to efficiently transfer knowledge across the entire dataset. Thus, to implement our efficient knowledge transfer approach, we employed Llama 3.1, a state-of-the-art open-source LLM.

Llama 3.1 represents a major upgrade over its predecessor, Llama 3, within the open-source LLM ecosystem. As an alternative to proprietary models, like GPT-3 and PaLM, Llama 3.1 provides transparency into model development while enabling responsible LLM utilization [30]. We selected Llama 3.1 to power our experiments, given its best-in-class performance among open-source LLMs, coupled with the permissive licensing for both academic work and commercial applications. Developed by Meta, Llama 3.1 is available in 8 billion, 70 billion, and 405 billion parameter configurations. The 70 billion parameters Llama 3.1 (Llama—3.1-70B) demonstrates state-of-the-art results on benchmarks like the MMLU and IFEval, surpassing proprietary models such as Llama 2-70B, Mixtral, and Falcon. Llama 3.1-70B shows similar abilities to major closed models like GPT-3.5 Turbo, GPT-4, Gemma 2, and PaLM-2-L. On the MMLU and GSM8K tests, Llama 3.1-70B performs better than

OpenAI's GPT-3.5 and has also improved on the programming tasks compared with Llama2-70B. Across almost all benchmarks, Llama 3.1-70B performs as well as or better than open-sourced LLMs [31].

Carefully following Llama's documentation,¹ we developed a prompt component to improve the efficiency of the prompts based on the official prompt template but also to allow the LLM to perform the topic modeling task fully unsupervised. The first part contains the role of the LLM. Creating prompts from the viewpoint of the role could enhance the relevance and appeal of the LLMs' responses. The benefits of role-based prompting include improving pertinence and exactness. Framing prompts around the mindset and context of the recipient helps the LLM better grasp the appropriate tone, level, and type of response expected. This additional role clarity and perspective-taking reduce inaccurate or off-base responses by grounding the model in the appropriate framing to determine relevant reactions, while the second part of the prompt includes instructions on what the LLM should do and what to return. We briefly experimented with different prompts and found the best-performing prompts to be:

- LLM Prompt : *You are a helpful, respectful and honest assistant for topic modeling. What is the topic of the following texts: D. Return only the topic.*

where, D represents the set of exemplar texts of the cluster used for the topic determination process.

4 Experimental results

This section presents an experimental validation of the proposed unsupervised topic modeling methodology using multiple benchmark datasets. To this end, we introduce a new metric leveraging the capabilities of LLMs to determine whether the unsupervised topic aligns with the original one. For reference, the performance of our approach is compared to BERTopic [15], a state-of-the-art topic modeling technique. BERTopic identifies latent topics within a text corpus by clustering with HDBSCAN at its core, making it a relevant baseline for comparison. In contrast, our method employs the PDDP algorithm, which is parameter-free, provides better control over the clustering structure, and retains balanced cluster sizes to help bound computational complexity. Furthermore, we perform a comparative analysis by directly prompting the LLM to discover topics for each text, referred to as the *LLM one-by-one* approach in the results table. The evaluation focuses on the precision of automatically discovered topics by measuring their alignment with ground truth labels after prompting the LLM with a fraction of texts per cluster. This tests the ability of our approach to rapidly adapt the LLM to new topics given just a few examples from each automatically induced cluster.

4.1 Benchmark datasets for validation

To validate the performance of our unsupervised topic modeling framework, we utilized multiple well-known annotated public datasets containing texts on various topics. These datasets, while not required for the unsupervised topic induction process itself, enable quantitative benchmarking against ground truth topic labels. The first dataset is the BBC News Articles dataset created by [32] for text categorization. This dataset contains 2,225 news articles published on the BBC website between 2004–2005, spanning 5 topical areas. The second dataset we leveraged was the 20 Newsgroups dataset,² containing 18,846 newsgroup documents divided into 20 unique categories. This dataset has become a widely used benchmark for training and testing Machine Learning models on text classification and clustering tasks. Additionally, we incorporated a dataset from Kaggle consisting of 20,006 abstracts of scientific articles from 4 disciplines: *Computer Science*, *Mathematics*, *Physics*, and *Statistics* [33]. We also included the E-commerce Reviews dataset comprising 50,425 texts across 4 e-commerce

¹ <https://ai.meta.com/llama/get-started/>

² <http://qwone.com/jason/20Newsgroups/>

domains, such as *Electronics*, *Housewares*, *Books*, and *Clothing/Accessories* [34]. Finally, we incorporated the “Bills” dataset containing 40,661 bill summaries from the 110–114th U.S. congresses with 21 high level human-annotated topics [19].

To ensure the analytical strength of our datasets, we implemented a comprehensive pre-processing procedure on our textual corpora before moving forward. This procedure involved the removal of non-lexical artifacts such as HTML formatting, whitespace, punctuation, and numeric strings. Moreover, we standardized all text to lowercase formatting and removed stop words, recognizing their tendency to introduce bias into aggregate statistics due to their high frequency and limited semantic content.

To evaluate our proposed framework, we first applied the PDDP algorithm to divide each benchmark dataset into clusters, as described in our methodology. To execute the PDDP algorithm, we utilized the HiPart package, which provides an open-source native Python implementation of efficient and interpretable divisive hierarchical clustering algorithms. This library enables interactive visualizations for manipulating the execution steps, allowing direct intervention in the clustering outcome. Due to a focus on computational efficiency in the implemented clustering methodologies, HiPart is highly applicable for large-scale data analytics applications [25].

The tunable parameters of the PDDP algorithm in the implementation of HiPart allow for the selection of the projection method, accommodating manifold spaces. However, since this was not necessary for our analysis, we used the default setting of PCA. For the other two parameters, we determined, based on the data and the task, that the maximum number of splits to perform on the data should be set at 600. This choice ensures significant segmentation of the data without losing too much coherence in the topic modeling. Additionally, the minimum number of samples required in each cluster is set to 20 to ensure minimal computational benefits while retaining enough samples to form a representative text.

After dividing our text corpora into clusters, we randomly selected 5 examples of texts from each cluster to form a representative sample. The decision to use a specific number of texts as representatives for each cluster was based on empirical trials, aiming for improved results while keeping time complexity and energy consumption as low as possible. The selection of texts is discussed further in Sect. 5. As follows, using these samples, we input queries into the LLM to induce a topic for each cluster. Finally, to use the pre-trained Llama 3.1-70B and execute queries, we deployed it locally using the vLLM framework [35] for optimized inference performance. The local machine utilized for this purpose was equipped with an AMD EPYC 7502 32-core CPU, 256GB of RAM, and two NVIDIA A100 GPUs with 80 GB of memory each. This high-performance hardware, combined with vLLM’s efficient serving capabilities, facilitated rapid querying of the Llama 3.1 model.

4.2 Evaluation metric

To validate the effectiveness of our unsupervised topic modeling approach, we developed an evaluation method that exploits the capabilities of LLMs. To the best of our knowledge, no existing evaluation metric has been established for comparing LLM-based generated topics with known ground truth topics. Our method allows for a quantitative assessment of the alignment between automatically discovered topics and original ground truth topics.

The evaluation process involves prompting an LLM (specifically, Llama 3.1) to determine which original labels best fit each discovered topic, providing a reliable metric of our framework’s performance. The validation process proceeds: For each topic identified by our proposed approach, we prompt the LLM with a carefully crafted query. The prompt structure was:

- Evaluation-Prompt: *Based on your previous answer: p_1 , which of the following topics fits the best? The topics are: T . Return only one topic from the provided list.*

where p_1 is the unsupervised topic of the cluster and T is the list of the original ground truth topics. To ensure the integrity of the evaluation, the LLM was reinitialized after each run of the proposed framework. This step was

crucial to prevent the model from retaining any information from previous evaluations, thus maintaining the independence of each evaluation. This method allowed us to quantitatively assess how closely our unsupervised approach could approximate the original topic structure. Figure 2 illustrates the schematic representation of this LLM-based evaluation process.

4.3 Performance evaluation

Regarding performance evaluation, Table 1 presents the outcomes of our framework alongside comparative results, showing the average score for each method over 10 independent iterations. For these experiments, all results for our Proposed Approach and the LLM one-by-one method were generated using 5 randomly selected representative samples per cluster, a choice made to ensure both robust evaluation and avoid bias. In each of the 10 independent iterations, these samples were chosen randomly, and the LLM was reloaded to prevent any prior knowledge retention.

Table 1 presents experimental results using the LLM-based evaluation metric. Our method achieved high alignment scores in most datasets while maintaining consistent efficiency. Specifically, we reached 0.861 on the BBC dataset, 0.820 on E-commerce, 0.784 on Papers, 0.694 on the Newsgroups dataset and 0.506 on the Bills dataset. Notably, the Newsgroups dataset initially led to underperformance due to its topic format (e.g., “rec.sport.hockey”). After simplifying these topics into more general categories (e.g., “sports”), which are more like category identifiers than human-interpretable topics, evaluation scores improved significantly, highlighting the LLM’s sensitivity to semantic coherence. Our method outperformed BERTopic on all datasets in terms of topic alignment. While the *LLM one-by-one* approach achieved the highest accuracy, it incurred a substantial computational cost, making it impractical for large-scale or time-sensitive applications.

To extend the evaluation of our proposed method, it is equally important to assess performance through established, widely used topic modeling and clustering metrics. These classic metrics offer complementary insights into topic coherence, diversity, and cluster-label correspondence, making them useful for benchmarking. To this end, we evaluated our method along with BERTopic and TopicGPT using both unsupervised metrics and supervised clustering metrics. Notably, we are using Topic Coherence (TC) [36], Topic Diversity [37], Adjusted Mutual Information (AMI) [38], Adjusted Rand Index (ARI), the Mutual Information Score (MIS), and the Harmonic Purity (HP). TopicGPT [19] is a recent prompt-based framework that assigns topics via individual LLM calls for each document. While the original implementation relies on closed-source LLMs such as GPT-4 for topic generation and GPT-3.5-turbo for assignment, we used the authors’ publicly available code and replaced the LLM backend with the same open-source model used in our framework to ensure a fair comparison. In their original experiments, the authors sampled only a small fraction of each dataset, ranging between 3% and 7%. We used a random 7% sample from each dataset to replicate this setup.

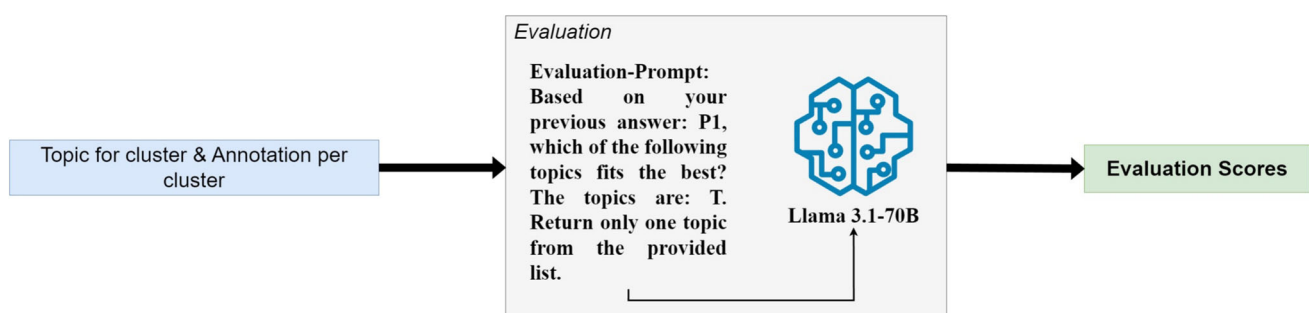


Fig. 2 Schematic representation of the LLM-based evaluation process for comparing generated topics with ground truth labels. The process involves using Llama 3.1-70B to assess the alignment between cluster-level topics and predefined ground truth topics, resulting in evaluation scores

Table 1 Experimental results of the proposed methodology. Evaluation score with standard deviation (indicated by $\pm$) and Δt (execution time in seconds) for each method

Dataset	Proposed approach		LLM one-by-one		BERTopic	
	Eval. Score	Δt	Eval. Score	Δt	Eval. Score	Δt
BBC	0.861 $\pm$ 0.009	505	0.963 $\pm$ 0.058	2,950	0.755 $\pm$ 0.06	216
Newsgroups	0.694 $\pm$ 0.015	474	0.983 $\pm$ 0.012	15,964	0.603 $\pm$ 0.02	1,744
E-commerce	0.820 $\pm$ 0.010	473	0.916 $\pm$ 0.085	47,613	0.736 $\pm$ 0.02	1,291
Papers	0.784 $\pm$ 0.003	474	0.956 $\pm$ 0.046	18,707	0.626 $\pm$ 0.01	1,466
Bills	0.506 $\pm$ 0.011	518	0.898 $\pm$ 0.016	74,417	0.431 $\pm$ 0.01	954

As shown in Table 2, our method performs competitively across all reported metrics. It achieves the highest overall scores on the BBC, E-commerce, and Bills datasets, especially excelling in clustering metrics such as AMI, MIS, and HP. While BERTopic performs slightly better on TC and TD for certain datasets, our approach demonstrates a more balanced performance between coherence and clustering quality. Notably, on more complex corpora like Newsgroups and Papers, our method maintains strong results without compromising efficiency. TopicGPT, although designed for semantic topic generation, shows inferior performance in clustering metrics, particularly in ARI and MIS, despite being evaluated on a significantly smaller subset of the data.

If we now turn our attention to runtime performance, our approach demonstrates significant reductions in execution time while sustaining strong topic discovery performance. As shown in Fig. 3, our method completes topic modeling in approximately 8 min across all datasets, maintaining remarkably consistent performance regardless of corpus size or complexity. In contrast, the *LLM one-by-one* approach scales poorly, requiring several hours to days on larger corpora such as E-commerce and Bills, despite its slightly higher accuracy. BERTopic exhibits moderate runtimes but remains slower than our approach. Finally, even though TopicGPT processes only a small sample of the data, it remains slower than both our method and BERTopic due to repeated document-level LLM interactions during topic assignment.

Table 2 Comparison of Topic Coherence (TC), Topic Diversity (TD), Adjusted Mutual Information (AMI), Adjusted Rand Index (ARI), Mutual Information Score (MIS) and Harmonic Purity (HP) across the proposed method, BERTopic and TopicGPT

Dataset	Method	TC	TD	AMI	ARI	MIS	HP
BBC	Proposed	0.549	0.733	0.645	0.675	1.078	0.831
	BERTopic	0.536	0.760	0.664	0.607	1.044	0.803
	TopicGPT	0.558	0.740	0.418	0.390	0.571	0.570
Newsgroups	Proposed	0.467	0.490	0.420	0.274	1.260	0.519
	BERTopic	0.524	0.750	0.416	0.276	0.689	0.684
	TopicGPT	0.409	0.579	0.270	0.120	0.436	0.325
E-commerce	Proposed	0.508	0.891	0.528	0.509	0.725	0.790
	BERTopic	0.503	0.808	0.435	0.410	0.612	0.731
	TopicGPT	0.401	0.538	0.316	0.221	0.374	0.505
Papers	Proposed	0.458	0.700	0.498	0.554	0.622	0.801
	BERTopic	0.468	0.725	0.521	0.550	0.622	0.816
	TopicGPT	0.467	0.545	0.391	0.296	0.395	0.601
Bills	Proposed	0.495	0.594	0.392	0.295	1.110	0.539
	BERTopic	0.509	0.552	0.346	0.130	0.924	0.467
	TopicGPT	0.443	0.415	0.237	0.054	0.342	0.266

The best performance for each metric is highlighted in bold

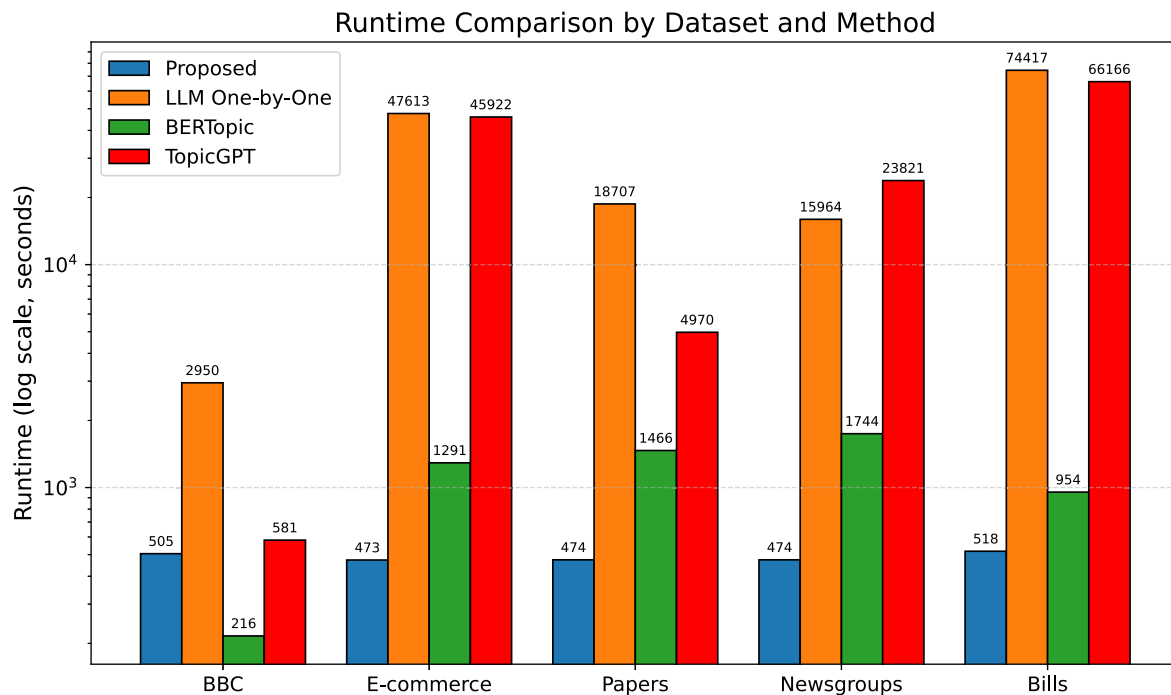


Fig. 3 Runtime comparison across datasets for the proposed method with five random samples, LLM One-by-One, BERTopic, and TopicGPT. The y-axis presents the seconds on a logarithmic scale

5 Discussion

5.1 Selection of representative texts

A critical aspect of our approach involved determining the optimal number of documents to process per cluster. We systematically experimented with using 1, 5, and 10 documents per cluster to evaluate the trade-offs between topic modeling performance, computational cost, and energy consumption. Our analysis, summarized in Table 3, ultimately led us to select 5 documents per cluster as the optimal balance.

Processing only a single document per cluster significantly reduced resource requirements but consistently led to noticeable performance degradation across all datasets. For instance, the BBC and Papers datasets experienced noticeable drops in performance, while the E-commerce, Newsgroups, and Bills datasets showed even more substantial decreases. These performance declines indicate that relying on a single representative document fails to capture the full semantic diversity within each cluster, resulting in less accurate topic characterization.

Conversely, analyzing 10 documents per cluster offered only marginal improvements in topic representation, and in some cases, even a slight decrease in performance (e.g., BBC and Bills datasets). More importantly, this approach incurred a substantially higher computational overhead. Execution times increased dramatically, with an

Table 3 Performance comparison across different numbers of representative samples per cluster. Evaluation scores with standard deviation and execution time differences (Δt) in seconds

Dataset	1 Sample		5 Samples		10 Samples	
	Score	Δt	Score	Δt	Score	Δt
BBC	0.853 ± 0.007	379	0.861 ± 0.009	505	0.866 ± 0.005	1063
Newsgroups	0.647 ± 0.092	384	0.694 ± 0.015	474	0.702 ± 0.008	779
E-Commerce	0.769 ± 0.021	340	0.820 ± 0.010	473	0.827 ± 0.012	783
Papers	0.754 ± 0.005	344	0.784 ± 0.003	474	0.784 ± 0.005	1356
Bills	0.386 ± 0.041	190	0.447 ± 0.011	518	0.425 ± 0.083	683

average increase of 444 seconds compared to the 5-sample configuration. This significant increase in processing time is primarily due to the much larger input token size when processing 10 documents concurrently, which creates computational bottlenecks during the LLM inference process. Although the 10-sample method theoretically has a slight advantage in capturing complex topic representations, its applicability is largely limited by time constraints and computational resources. The marginal performance gains may outweigh the higher cost in situations where processing time is not a crucial consideration and computational power is available.

Our comprehensive trade-off analysis reveals that the 5-sample configuration is the optimal sweet spot. Transitioning from 1 to 5 samples yielded significant performance gains with only a moderate increase in execution time. In contrast, increasing the sample size from 5 to 10 documents provided minimal, or even negative, performance changes while drastically escalating computational demands. This strongly suggests that 5 representative documents per cluster effectively capture sufficient semantic diversity for accurate topic modeling without incurring the exponential computational costs associated with larger sample sizes.

An essential design decision in our framework is the intentional use of over-clustering. This approach ensures that semantically distinct documents are more likely to be separated early in the process, reducing intra-cluster topic variance and enabling the LLM to generate more precise topics from representative samples. Over-clustering also mitigates the risk of assigning a single, generic topic to a heterogeneous group of documents, which is a common failure mode in under-clustered systems. This approach is consistent with results from the literature, especially from research on recursive partitioning techniques like PDDP [39]. These techniques show that internal cohesion and semantic noise are generally improved by increasing cluster granularity. Moderate over-clustering produces more semantically coherent groupings and makes downstream topic inference more efficient, whereas excessive splitting can ultimately result in fragmentation or redundancy.

In our implementation, we define a maximum number of allowed splits, thereby restricting the depth of recursive clustering. Our experiments demonstrate that small adjustments to this threshold have little to no impact on clustering quality or computational cost, and that this value functions as a practical cutoff rather than a tuned hyperparameter. The process would deteriorate to the point where each sample is assigned to its own cluster, removing any meaningful grouping, if the number of splits is equal to the number of documents. In reality, we discover that a threshold of a few hundred is suitable to produce good outcomes on a variety of datasets. This degree of over-clustering strikes a good balance between enhancing semantic coherence and enhancing the reliability of downstream topic inference without causing fragmentation or redundancy.

5.2 Qualitative findings and topic resolution

The qualitative assessment of the topic modeling results produced by the LLM on various benchmark datasets revealed several instances where the output of the model appeared to surpass the accuracy of the provided ground truth labels. Specifically, the analysis highlights two notable cases: the “BBC” dataset and the “Papers” dataset.

In the “BBC” dataset, the LLM identified the primary topic of a passage concerning a UK game company facing a financial crisis as “business,” in contrast to the ground truth label of “tech.” Despite the involvement of the company in the tech industry, the passage’s emphasis on the financial and operational challenges; such as bankruptcy and layoffs, makes “business” a more appropriate and insightful topic label. This suggests that the LLM was able to capture the underlying focus of the text more effectively than the existing ground truth.

Similarly, in the “Papers” dataset, the topic identification of the LLM labeled a paper about “Vizic” a Python visualization library for observational astronomy, as “computer science,” whereas the ground truth labeled it as “Physics”. The passage’s primary content revolves around the development and application of a software tool for data visualization and analysis, which aligns more closely with computer science. While the context of the tool’s application is related to astronomy, the core discussion centers on aspects of software development and data processing, thus making “computer science” a more precise topic.

These findings underscore the potential of the LLM to generate more refined and contextually accurate topic labels compared to the ground truth, particularly in cases where the existing labels may overlook the specific

focus or dominant theme of the text. The supplementary materials provided further detail and examples from each dataset, reinforcing the observation that the approach of the LLM to topic modeling can sometimes reveal insights or connections that are not captured by the ground truth labels. This qualitative evaluation demonstrates the added value of the LLM in refining and enhancing topic classification beyond conventional labeling methods.

5.3 Prompting strategies and cross-model validation

To investigate the stability and reliability of the proposed LLM-based evaluation method, we conducted a prompt ablation study using the same Llama 3.1–70B model across all datasets. Four prompt designs were evaluated:

- Minimalist: *Given the topic: p_1 , choose the best matching topic from: T .*
- Chain-of-Thought (CoT): *You are a helpful assistant for identifying topics. First, analyze the given text: p_1 . Then, among the following topics: T , select the most appropriate one. Return only one topic.*
- Multi-choice (MC): *You are a helpful assistant. Here is a topic: p_1 . Select the best matching topic from the following options: T . Only return one label.*
- Explicit Relevance Scoring (ER): *You are an assistant for evaluating topic relevance. Given the text: p_1 , review the following topics: T . Select the topic most semantically relevant to the text. Return only one topic.*

Each design was applied independently across all datasets, and we recorded evaluation scores and standard deviations. Although some variants, particularly Multi-choice, occasionally achieved scores comparable to our proposed prompt (described in Sect. 4.2 and used as the baseline), the performance varied considerably depending on dataset characteristics. Notably, the Chain-of-Thought prompt underperformed in domains with long or information-dense texts, suggesting an increased susceptibility to verbosity and token drift. In contrast, our proposed prompt consistently achieved either the highest or second-highest performance across all datasets, with minimal variance. These results confirm that the structure of the prompt itself can significantly affect evaluation outcomes, even when using the same language model and data. The proposed prompt serves as the baseline for comparison in Fig. 4.

The proposed framework employs Llama 3.1-70B for both topic induction and assessment. Although this integrated approach shows considerable efficiency, it may introduce potential evaluation bias since depending on the same model for both phases might favor its generation tendencies. To overcome this limitation and verify the reliability of our results, we conducted an assessment utilizing several independent cutting-edge language models. In particular, we conducted the evaluation process again using DeepSeek-R1:70B, Qwen3:32B, Qwen2.5:32B, and Gemma2:27B, along with Llama3.3, which is an independent checkpoint within the Llama series.

Figure 5 displays the findings from this comparative analysis involving all datasets. The visual representations illustrate the disparity in evaluation scores obtained by each model concerning the original Llama 3.1-70B baseline. Importantly, in every instance, the variations in performance stay quite minor, with none of the other models showing consistently greater alignment scores. Specifically, Llama3.3 yielded similar or marginally enhanced scores, strengthening the credibility of the assessment even among the Llama model lineage. Conversely, DeepSeek-R1:70B and Qwen3:32B showed somewhat reduced alignment, particularly on the E-Commerce and Newsgroups datasets. Nonetheless, these variations did not notably affect the overall conclusions. This cross-model validation offers further proof that the evaluation findings in this study are not excessively reliant on Llama 3.1-70B, but hold strong across a variety of competitive language models.

5.4 Efficiency and comparative advantages

The proposed framework demonstrates key improvements in both energy and computational efficiency compared to the existing methods. By utilizing computationally inexpensive hierarchical clustering and strategically selecting a fraction of cluster-representative documents, we avoid costly inference time and resource usage associated with passing every corpus through the LLM. This reduction in LLM queries translates into substantial

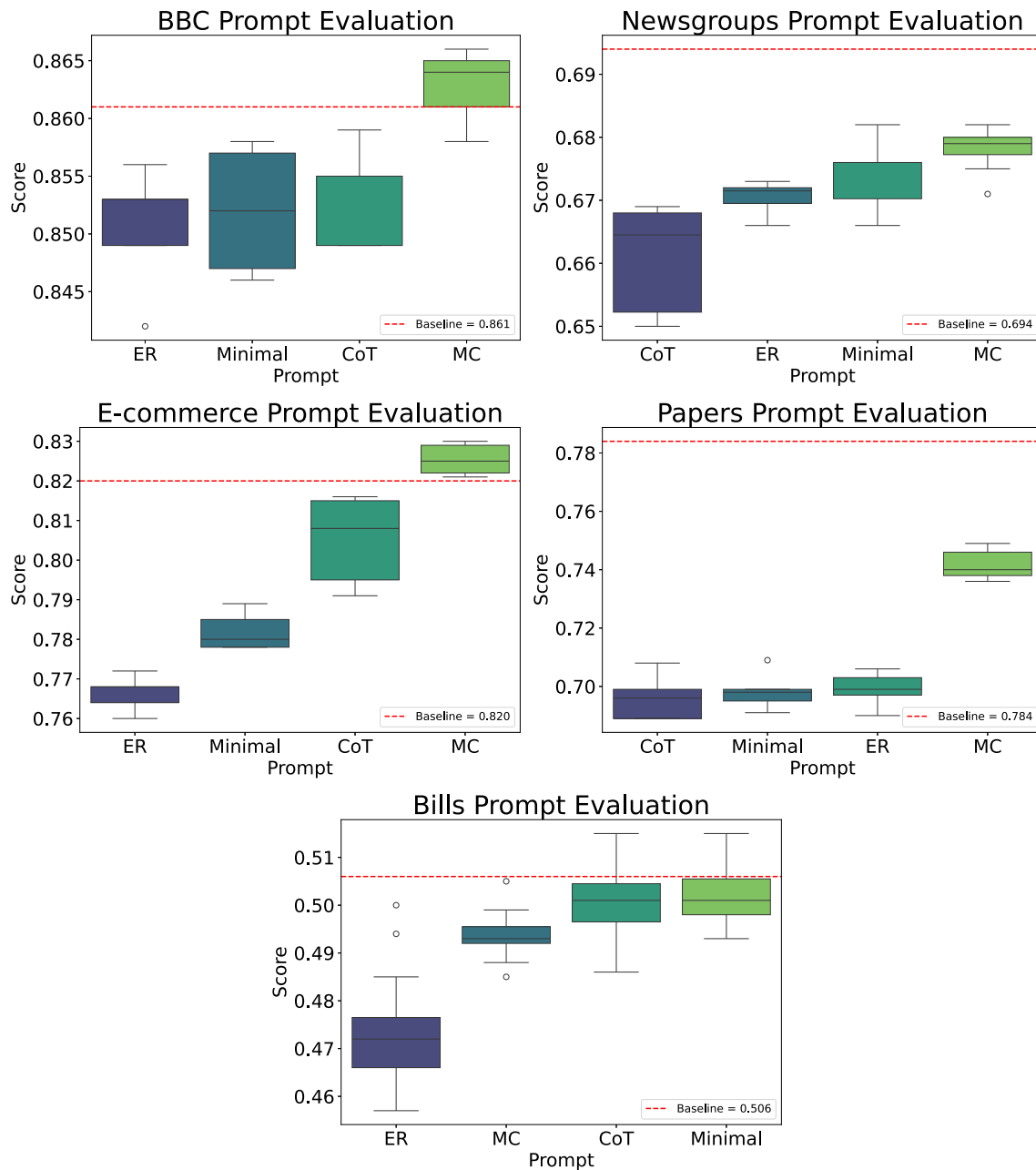


Fig. 4 Evaluation score of different prompt designs across all datasets. Each bar shows the average performance of a given prompt: Minimalist, Chain-of-Thought (CoT), Multi-choice (MC), Explicit Relevance Scoring (ER), when used with Llama 3.1–70B. The red dashed line indicates the performance of the baseline prompt used in our main experiments

decreases in energy consumption and operating costs. We estimate our approach decreases energy usage up to 98% depending on the dataset size. Table 4 presents the energy consumption in kWh and running costs in EUR for each method. The running costs have been estimated on the average €/kWh in the European Union for July 2024, before tax, at a rate of €0.2416/kWh. Total consumption values are not included in the evaluation process of each method.

A key architectural advantage of our approach is that execution time and energy consumption are primarily determined by the number of clusters formed, rather than the total dataset size. This is because computational cost scales with the number of LLM inference calls, which in our method depends on the clustering structure and the

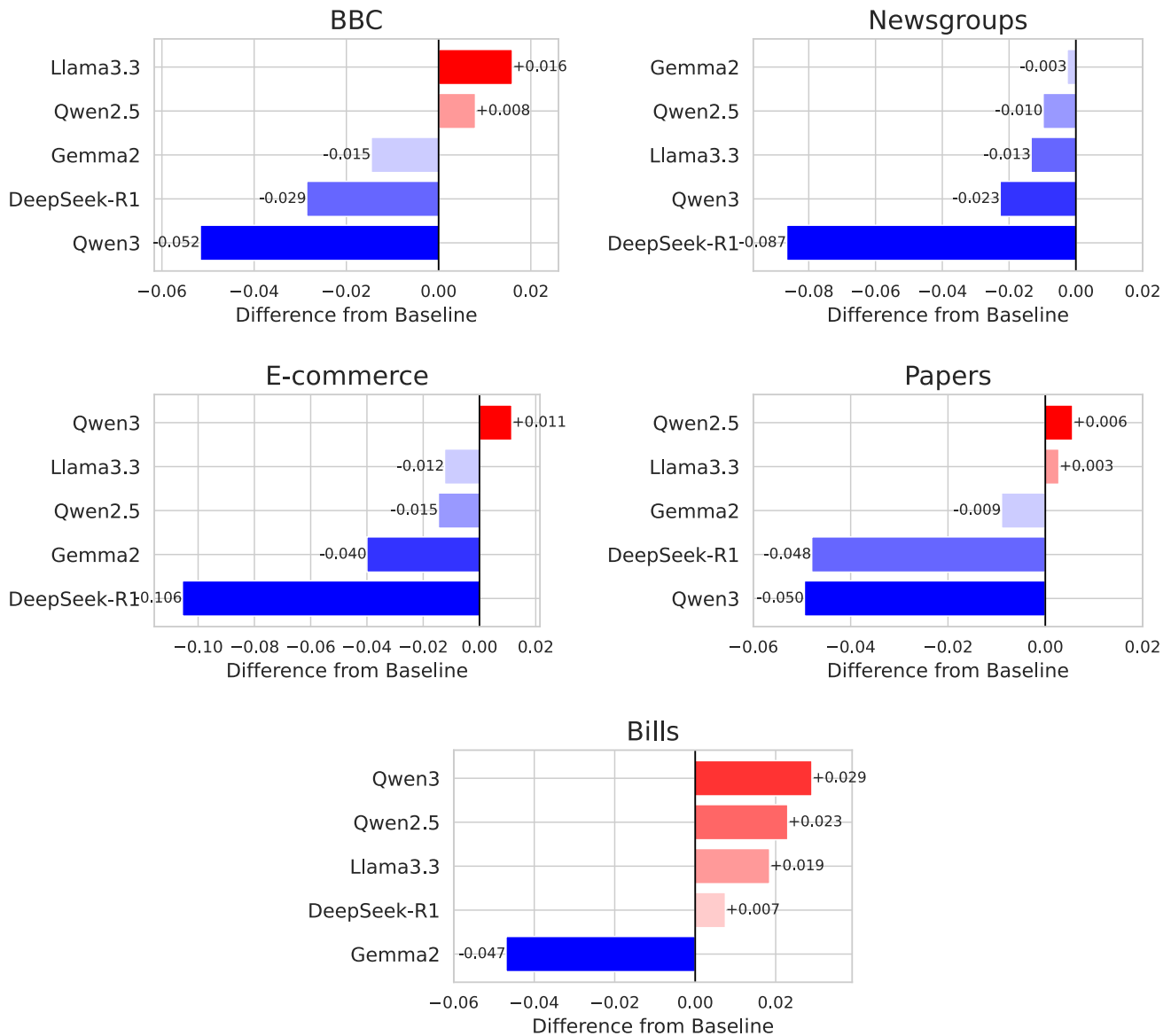


Fig. 5 Cross-model evaluation results across all benchmark datasets. The plots illustrate the difference in evaluation scores produced by alternative state-of-the-art LLMs compared to the original Llama 3.1-70B baseline. Positive values (with red shading) indicate improved alignment, while negative values (with blue shading) reflect lower agreement

Table 4 Detailed comparison of the energy consumption in kWh and running costs in EUR associated with each approach

Dataset	Proposed Approach		LLM one-by-one		BERTopic		TopicGPT	
	kWh	Run cost (€)	kWh	Run cost (€)	kWh	Run cost (€)	kWh	Run cost (€)
BBC	0.10	0.024	0.29	0.07	0.01	0.002	0.05	0.012
Newsgroups	0.06	0.014	2.17	0.507	0.24	0.058	1.29	0.311
E-commerce	0.06	0.014	4.53	1.087	0.19	0.046	2.54	0.614
Papers	0.07	0.017	2.39	0.555	0.21	0.051	0.46	0.112
Bills	0.06	0.014	4.12	0.991	0.14	0.034	3.80	0.919

The lowest running cost for each approach is highlighted in bold

number of representative samples per cluster. By bounding the maximum number of splits and fixing the number of exemplars, we effectively control and limit the total number of LLM interactions, keeping costs proportional to the clustering structure rather than corpus size. This stands in contrast to methods like TopicGPT and LLM one-by-one inference, where costs scale linearly with dataset size.

Our cluster-based approach achieves dramatic efficiency gains compared to methods that process each document individually through LLMs. Against LLM one-by-one, the "Papers" dataset, we reduced energy consumption by 96%, on the "E-commerce" dataset by 98%, and on the "Newsgroups" dataset by 97%. The BBC dataset showed a more moderate but still substantial 65% reduction, likely due to its smaller corpus size compared to other datasets. Compared to TopicGPT, our method demonstrates similar energy savings. Notably, despite TopicGPT processing only a fraction of the data (7%), it still consumes significantly more energy than our approach due to its document-level LLM interactions.

When comparing our approach to BERTopic, we observe some interesting trade-offs. While our method consistently reduces energy consumption across most datasets, the extent of the savings varies. On the "Newsgroups", "E-commerce" and "Papers" datasets, we achieved energy reductions of 75%, 68%, and 66%, respectively. However, on the "Bills" dataset, the savings were more moderate at 57%. Interestingly, on the "BBC" dataset, our approach consumed more energy than BERTopic, but this was offset by a 14% improvement in topic modeling performance.

The observed differences in energy consumption are primarily driven by the number of clusters produced during the topic modeling. Our method generated a stable number of clusters across datasets while BERTopic exhibits greater variability. In some cases, BERTopic produced extremely high numbers of clusters, 1,714 on the E-Commerce dataset and 1,000 on the Bills. Moreover, we found that it did not consistently produce the same number of clusters when run multiple times on the same dataset, further contributing to unpredictable energy usage. Crucially, despite generating significantly more clusters, BERTopic often achieved lower topic modeling accuracy compared to our method.

5.5 Summary of framework advantages

To further highlight the distinctiveness of our framework, we summarize here the key contributions and comparative advantages over existing state-of-the-art approaches. In summary, the proposed framework exhibits several distinctive advantages compared to existing methods. First, by combining divisive hierarchical clustering (PDDP) with selective LLM-based topic discovery, it achieves significant reductions in computational time and energy consumption, as evidenced by runtime improvements of up to 97% and energy savings up to 98% compared to the LLM one-by-one baseline. Second, unlike BERTopic, which uses density-based clustering and transformer embeddings, our approach leverages a deterministic, parameter-efficient clustering process that better controls the granularity of topic formation. Third, the introduction of an LLM-based evaluation strategy provides a novel mechanism for validating unsupervised topic modeling outputs against ground truth, enhancing reliability without requiring labeled datasets during training. These innovations collectively establish our method as a more scalable, sustainable, and effective alternative for large-scale topic modeling tasks.

5.6 Limitations

Although the efficiency and scalability of LLM-based topic modeling are greatly improved by our suggested framework, it's crucial to recognize some inherent drawbacks resulting from its fundamental design ideas. First, the quality and structure of the initial hierarchical clustering are closely related to our method's effectiveness and efficiency. An important architectural advantage of our approach is that its computational cost scales mainly with the number of clusters, not the size of the entire corpus. However, this also implies that the clustering output directly determines the topic granularity. Because the LLM's inference is based on a small number of representatives, an excessively coarse clustering, while optimizing efficiency, runs the risk of oversimplifying diverse

topics within large clusters. On the other hand, an excessively fine-grained clustering may result in extra LLM calls, which could decrease the overall efficiency gains. To balance topic granularity with computational cost, it can be necessary to carefully empirically adjust clustering parameters, such as the number of exemplars and the maximum splits. The second foundation of our framework is that the chosen cluster-representative documents completely indicate the semantic core of their respective clusters. Although this is usually the case for well-formed clusters, documents that are true outliers or are part of very small, isolated clusters may find this method problematic. The typical selection procedure may not fully represent these documents, which could result in less accurate topic discoveries for them.

6 Conclusion

In this work, we have presented an energy and time-efficient framework for unsupervised topic modeling of large corpora. Our key novelty is a two-stage approach that first exploits computationally inexpensive Hierarchical Divisive Clustering. We then sample a small fraction of exemplars from each distinct cluster and take advantage of language models to understand topics at the cluster level. By transferring the cluster-level topics to all documents, we can effectively annotate large corpora without the need to pass each document through the LLM. Compared to directly modeling topics with LLMs on the full corpus, our framework exhibits comparable results but yields major improvements in time and energy efficiency. Moreover, the proposed framework has much lower power consumption and execution time compared to the state-of-the-art BERTopic approach. Looking ahead, the next steps of this research lead to the execution of more specialized experiments in not only benchmark datasets, but real-world datasets, as well. Also, it is important to investigate the automation of the number of cluster selection processes, along with the number of exemplar samples from each cluster. This will alleviate the need for fine-tuning the parameters of the algorithm, further lowering its energy consumption. Finally, as the use of LLMs is spreading in various fields for various tasks, it is also important to study the viability of the framework in other fields.

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Data Availability The data used in this paper are publicly accessible, and details are available in the references section.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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